

1. Administrative & Study Identification

Full Study Title: A Phase III, Multicenter, Randomized, Visual Assessor-masked, Active-comparator Study of the Efficacy, Safety, and Pharmacokinetics of the Port Delivery System With Ranibizumab in Chinese Patients With Neovascular Age-related Macular Degeneration **Study Title:** A Study of the Efficacy, Safety, and Pharmacokinetics (PK) of the Port Delivery System With Ranibizumab (PDS) in Chinese Participants With Neovascular Age-related Macular Degeneration (nAMD) **Protocol Number:** YR42983 **NCT Number:** NCT05562947 **Trial Status:** Recruiting **Sponsor/Company Name:** Roche (Hoffmann-La Roche) **Investigational Product:** Port Delivery System (PDS) with Ranibizumab **Phase of Development:** Phase III **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy, safety, and pharmacokinetics (PK) of ranibizumab 100 mg/mL delivered every 24 weeks (Q24W) via the PDS implant compared with ranibizumab 0.5 mg delivered every 4 weeks (Q4W) as an intravitreal (IVT) injection in Chinese participants with nAMD. **Secondary Objectives:** To assess the safety profile of the PDS (including rates of vitreous hemorrhage and endophthalmitis), monitor steady-state PK concentrations, and evaluate long-term visual and anatomic outcomes.

2.2 Background & Rationale To address the high treatment burden of frequent (monthly or bimonthly) intravitreal injections required for patients with neovascular age-related macular degeneration (nAMD). The Port Delivery System (PDS) is designed to provide continuous delivery of ranibizumab to the vitreous, maintaining therapeutic drug concentrations for at least 24 weeks, thereby reducing the number of clinical visits and injections while aiming for comparable clinical and visual outcomes to standard monthly injections.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Ranibizumab 0.5 mg IVT injection) **Blinding:** Visual Assessor-masked **Randomization:** Randomized

3.2 Planned Enrollment & Duration Target \$N\$: 68 participants
Number of Sites: Multicenter (in China) **Estimated Study Start:** 17-Jun-2024 **Estimated Primary Completion:** 2029

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Initial diagnosis of nAMD within 9 months prior to the screening visit. 2. Previous treatment with at least three anti-vascular endothelial growth factor (VEGF) IVT injections for nAMD per standard of care within 6 months prior to the screening visit. 3. Demonstrated response to prior anti-VEGF IVT treatment since diagnosis. 4. Availability of historical VA data prior to the first anti-VEGF treatment for nAMD up to the screening visit. 5. Best-Corrected Visual Acuity (BCVA) of 34 letters or better (20/200 or better approximate Snellen equivalent), using the Early Treatment Diabetic Retinopathy Study (ETDRS) chart at a starting distance of 4 meters. 6. All subtypes of nAMD lesions are permissible. 7. Sufficiently clear ocular media and adequate pupillary dilation to allow for analysis and grading by the central reading center of FP, FA, ICGA, FAF, and OCT images.

4.2 Exclusion Criteria Prior Ocular Treatment (Study Eye): 1. History of vitrectomy surgery, submacular surgery, or other surgical intervention for AMD. 2. Prior treatment with Visudyne, external-beam radiation therapy, or transpupillary thermotherapy. 3. Previous treatment with corticosteroid IVT injection or previous intraocular device implantation (excluding IOL implants). 4. Previous laser (any type) for AMD treatment, or prior intravitreal treatments for geographic atrophy. 5. Treatment with anti-VEGF agents other than ranibizumab within 1 month prior to the randomization visit. 6. Concurrent conjunctival, Tenon's capsule, and/or scleral condition in the supero-temporal quadrant that may affect PDS implantation, tissue coverage, or refill procedures.

Prior Treatment (Either Eye): 1. Prior treatment with brodalumab or prior gene therapy for nAMD or other ocular diseases. 2. Previous participation in any ocular disease studies of investigational drugs/devices within 3 months (or five elimination half-lives).

Lesion Characteristics & Concurrent Conditions: 1. Subretinal hemorrhage involving the center of the fovea (if >0.5 disc area / 1.27 mm^2), subfoveal fibrosis, or subfoveal atrophy in the study eye. 2. CNV due to other causes (e.g., ocular histoplasmosis, trauma, pathologic myopia) in either eye. 3. Retinal pigment epithelial tear, active intraocular inflammation, history of vitreous hemorrhage, rhegmatogenous retinal detachment/tears, or pars plana vitrectomy in the study eye. 4. Aphakia, absence of posterior capsule, or refractive error demonstrating >8 diopters of myopia in the study eye. 5. Intraocular surgery within 3

months, uncontrolled ocular hypertension/glaucoma, or corneal transplant in the study eye. 6. Non-functioning fellow eye; active infectious conjunctivitis, keratitis, scleritis, endophthalmitis, or history of uveitis in either eye.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Experimental Arm:** PDS implant surgically inserted and filled with approx. 20 µL of 100 mg/mL ranibizumab (approx. 2 mg dose). Implant refill-exchanges occur at fixed 24-week intervals (Q24W). * **Active Comparator Arm:** Intravitreal (IVT) injections of 50 µL of 10 mg/mL ranibizumab (0.5 mg dose) delivered every 4 weeks (Q4W). **Route of Administration:** Surgically implanted intraocular Port Delivery System (PDS) or Intravitreal (IVT) Injection. **Duration of Treatment:** Long-term continuous delivery with fixed refills over the study duration (expected up to 240 weeks).

5.2 Concomitant & Prohibited Therapy Permitted: Standard of care treatments for non-ocular co-morbidities. **Prohibited:** Treatment with anti-VEGF agents other than ranibizumab within 1 month prior to randomization, prior corticosteroid IVT injections, or concurrent participation in other investigational drug/device studies.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, BCVA Assessment, Retinal Imaging, Medical History
Baseline	Day 1	Randomization, PDS Implantation (Experimental) or 1st IVT Injection (Comparator)
Interim	Weeks 4 to 24	IVT arm receives Q4W injections; PDS arm monitored for safety/BCVA until first Q24W refill
End of Study	Week 240 (Approx)	Final BCVA, Retinal Imaging, Exit Interview, Safety Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Change From Baseline in Best-corrected Visual Acuity (BCVA) Score Averaged Over Weeks 36 and 40. **Measurement Method:** Visual Assessor-masked evaluation using the Early Treatment Diabetic Retinopathy Study (ETDRS) Visual Acuity (VA) Chart at a starting distance of 4 meters.

7.2 Secondary & Exploratory Endpoints * Change From Baseline in BCVA Score Over Time. * Proportion of Participants With BCVA Score of 38 Letters (20/200 Approximate Snellen Equivalent) or Worse Averaged Over Weeks 36 and 40. * Proportion of Participants With BCVA Score of 38 Letters (20/200 Approximate Snellen Equivalent) or Worse Averaged Over Weeks 44 and 48. * Incidence

and severity of ocular and non-ocular adverse events (e.g., endophthalmitis, vitreous hemorrhage). * Change from baseline in central foveal thickness (CFT) via optical coherence tomography (OCT). * Steady-state pharmacokinetics of ranibizumab in serum and aqueous humor.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via regular study visits. Special focus on procedure-related complications like implant dislocation, conjunctival retraction, vitreous hemorrhage, and endophthalmitis. **Serious Adverse Events (SAEs):** Must be reported within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** Independent external data monitoring committee oversight to ensure participant safety.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy non-inferiority comparisons; Safety Set for all AE/SAE evaluations. **Sample Size Justification:** $N=68$ participants, appropriately powered to determine the non-inferiority (using standard non-inferiority margins such as ± 4.5 ETDRS letters) of the PDS Q24W regimen compared to the IVT Q4W regimen. **Handling of Missing Data:** Last Observation Carried Forward (LOCF) and Mixed-Effect Model Repeated Measure (MMRM) for continuous longitudinal efficacy endpoints (e.g., BCVA changes).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring visits to ensure surgical protocol adherence and masking integrity. **Audit Trail:** Verified eCRF locking, centralized reading center for objective OCT grading, and independent visual assessors.

11. Study Identifiers & Governance

Posting ID: {{312495241994001045}} **Primary ID:** {{YR42983}}
Registry Number: {{NCT05562947}} **Sponsor/Lead Organization:** {{Roche / Hoffmann-La Roche}} **Study Title:** {{A Study of the Efficacy, Safety, and Pharmacokinetics (PK) of the Port Delivery System With Ranibizumab (PDS) in Chinese Participants With Neovascular Age-related Macular Degeneration (nAMD)}} **Official Regulatory Title:** {{A Phase III, Multicenter, Randomized, Visual Assessor-masked, Active-comparator Study of the Efficacy, Safety, and Pharmacokinetics of the Port Delivery System With Ranibizumab}}

in Chinese Patients With Neovascular Age-related Macular Degeneration}} **Trial Status:** {{Recruiting}}

12. Clinical Framework (Study Specific)

Disease Areas: {{Neovascular Age-related Macular Degeneration, nAMD}} **Primary Condition:** {{Neovascular Age-related Macular Degeneration (nAMD)}} **Diagnosis Threshold:** {{nAMD diagnosis with response to prior anti-VEGF IVT and BCVA $\geq$ 34 letters}} **Medical Methodology:** {{Anti-VEGF therapy via Port Delivery System (PDS) or Intravitreal (IVT) Injection}} **Optimization Target:** {{Maintenance of visual acuity and reduction of treatment burden}}

13. Study Procedures & Methodology

Management Pathway: {{Surgical implantation and fixed-interval refill-exchanges for the PDS arm vs. Q4W IVT injections}} **Education Protocol (HCP):** {{Surgical training for pars plana PDS implantation and safe refill-exchange procedure}} **Education Protocol (Patient):** {{Post-operative ocular care and prompt identification of endophthalmitis/hemorrhage symptoms}} **Quality Audit Mechanism:** {{Strict visual assessor masking and central reading center for anatomical measurements}}

14. Observation & Follow-Up Schedule

Total Duration: {{Approx. 240 Follow-up weeks}} **Onsite Visit Cadence:** {{Every 4 weeks (Q4W) for IVT; Every 24 weeks (Q24W) for PDS refills}} **Remote Monitoring Frequency:** {{As needed for safety evaluations and AE tracking}} **Checklist Review Interval:** {{Prior to each scheduled IVT injection or surgical refill procedure}}

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	____	[DD-MMM-YYYY]				